

Marisa Schmidt Bazzi, Ph.D.

Postdoctoral Scholar, Cardiovascular Biomechanics Computation Laboratory
Stanford University, Palo Alto, CA, USA

msbazzi@stanford.edu
+1 (612) 443-5900
marisabazzi.com

Research Interests

Computational mechanobiology of cardiovascular adaptation, integrating subject-specific fluid–solid interaction, constrained-mixture growth and remodeling, multimodal imaging, and data-driven constitutive modeling.

Education

Ph.D. in Materials Science, University of Minnesota–Twin Cities

Sep 2018–Jan 2024

Minneapolis, MN

Thesis: Multiscale, Multiphysical Phenomena in Arterial Remodeling.

Advisor: Victor H. Barocas.

M.Sc. in Mechanical Engineering, Pontifical Catholic University of Rio de Janeiro

Jan 2016–Jan 2018

Rio de Janeiro, Brazil

Dissertation: Numerical investigation of viscoelastic liquid curtain breakup.

Advisor: Marcio S. Carvalho.

B.Sc. in Mechanical Engineering, Pontifical Catholic University of Rio de Janeiro

Jan 2010–Dec 2015

Rio de Janeiro, Brazil

Undergraduate thesis: Modeling of asphaltene deposition.

Research Experience

Postdoctoral Scholar, Stanford University

Feb 2024–Present

Cardiovascular Biomechanics Computation Laboratory, Palo Alto, CA

- Lead computational modeling efforts to understand and predict tissue-engineered vascular graft (TEVG) remodeling by integrating longitudinal imaging, scaffold mechanics, fluid–structure interaction, and constrained-mixture growth and remodeling.
- Develop physics-constrained neural-network approaches to discover interpretable constitutive laws for vascular tissues, with applications to venous mechanics.
- Lead experimental–computational studies of developmental changes in inferior vena cava geometry, hemodynamics, and mechanics in a juvenile large-animal model.
- Develop computational functionality within the open-source SimVascular framework for cardiovascular fluid–solid–growth simulations.

Graduate Research Assistant, University of Minnesota–Twin Cities

Sep 2018–Jan 2024

Department of Biomedical Engineering, Minneapolis, MN

- Developed subject-specific computational models to investigate how altered blood rheology in sickle cell disease affects systemic hemodynamics and oxygen transport in the cerebral vasculature.
- Built image-based fluid–structure interaction models of ascending thoracic aortic aneurysms to identify biomechanical markers associated with disease progression and outcomes.
- Developed multiscale growth-and-remodeling models linking vascular wall mechanics to longitudinal changes in aneurysm geometry and arterial adaptation.
- Integrated medical imaging, experimental measurements, finite-element modeling, and transport simulations to study cardiovascular disease across multiple spatial and temporal scales.

Research Scientist, PUC–Rio

Jan 2018–Sep 2018

Laboratory of Microhydrodynamics and Flow in Porous Media, Rio de Janeiro, Brazil

- Developed mathematical and computational models of two-dimensional multiphase flow in porous media for reservoir-

flow and pressure analyses.

- Implemented numerical simulations to investigate fluid compressibility and reservoir response under varying pressure conditions.
- Applied finite-element and numerical methods to characterize transport processes in complex porous-media systems.

Graduate Research Assistant, Pontifical Catholic University of Rio de Janeiro

Jan 2016–Jan 2018

Laboratory of Microhydrodynamics and Flow in Porous Media, Rio de Janeiro, Brazil

- Developed mathematical models of viscoelastic liquid-sheet and liquid-curtain rupture to determine how non-Newtonian stresses regulate interfacial stability.
- Derived long-wave models for planar and axisymmetric geometries and implemented numerical methods to solve the resulting governing equations.
- Performed linear stability analyses demonstrating how viscoelastic normal stresses can delay or suppress liquid-sheet rupture.

Funding and Awards

CHIP T32 Fellowship	2026–Present
Rising Star in Computational and Data Science	2025
Baker Postdoctoral Fellowship	2024–2025
Equity and Inclusion Leadership Showcase	2023
President’s Student Leadership & Service Award	2023
American Heart Association Predoctoral Fellowship Ranked in the top 8%.	2022–2024
CEMS Teaching Assistant Award	2021
SB3C Diversity Travel Grant	2019
Second Place, ABCM–EMBRAER Best M.Sc. Dissertation	2018
FAPERJ Nota 10 Fellowship for M.Sc. Program	2017–2018

Teaching and Mentoring Training

Postdoc Mentoring Certificate – Tier 1: Learn , Stanford University Office of Postdoctoral Affairs <i>Mentoring Development Program</i>	2026
• Completed foundational mentor training, structured mentoring discussions, self-assessment, and reflective development of mentoring practices.	
Optimizing Culturally Inclusive Mentoring Practices , Stanford University Office of Postdoctoral Affairs <i>Two-Day Intensive Mentoring Workshop</i>	2025
• Completed intensive mentor training led by Kelly Diggs-Andrews, Ph.D., focused on effective and culturally responsive research mentoring.	
Teaching Workshop for Postdocs , Stanford University Office of Postdoctoral Affairs <i>Teaching Development Program</i>	2025
• Completed formal training in evidence-based teaching, including learning climate, communication of goals, learner assessment, feedback, and promotion of self-directed learning.	
Postdoc Mentoring Coach Training , Stanford University Office of Postdoctoral Affairs <i>Peer Mentoring Coach and Facilitator Training</i>	2025
• Trained in CIMER-based research mentoring, peer coaching, mentoring curriculum facilitation, and workshop facilitation.	

Teaching Experience

- Introduction to Bioengineering Research**, Stanford University 2025
Guest Lecturer
- Delivered a guest lecture on approaches to computational modeling and simulation research in congenital heart disease.
- Biomedical Engineering Undergraduate Seminar II**, University of Minnesota 2023
Guest Lecturer
- Taught a module on the connections between sickle cell disease (SCD) and cerebral aneurysms.
- Numerical Methods for Materials Science**, University of Minnesota 2022
Teaching Assistant
- Held office hours and led discussion sections to support students in applying numerical methods to materials science problems.
- Materials Processing**, University of Minnesota 2020
Teaching Assistant
- Led laboratory sessions on thermal and mechanical processing methods used to control material structure and properties.
- Introduction to the Science of Engineering Materials**, University of Minnesota 2019
Teaching Assistant
- Led laboratory sessions testing to analyze properties like elasticity, fracture toughness, and thermal behavior.
- Numerical Methods**, Pontifícia Universidade Católica do Rio de Janeiro 2016–2018
Substitute Instructor
- Delivered undergraduate lectures in place of the course instructor, covering numerical integration, Newton methods, and other fundamental numerical techniques.

Research Supervision

- D. A.**, Stanford University 2024–Present
Graduate Researcher
- Applied constitutive artificial neural networks (CANNs) to discover interpretable constitutive models for coronary artery bypass grafts.
 - Assessed the influence of constitutive model choice on fluid–structure interaction (FSI) predictions.
- L. W.**, Stanford University 2025
Undergraduate Researcher
- Performed longitudinal analysis of large-animal tissue-engineered vascular graft remodeling using high-fidelity FSI simulations.
- C. A.**, The Ohio State University 2024–2025
Graduate Researcher
- Analyzed longitudinal vascular geometry and mechanics in a large-animal model of venous development.
 - Performed quantitative analysis of longitudinal imaging and biomechanical data from tissue-engineered vascular graft studies.
 - Subsequently admitted to medical school at Ohio State University.
- C. L.**, Stanford University 2024
Undergraduate Researcher
- Performed patient-specific FSI simulations of Fontan cardiovascular models.
 - Subsequently admitted to the Stanford Bioengineering graduate program.
- J. F.**, University of Minnesota 2022–2023
Graduate Researcher
- Applied signal-processing methods to study pressure waveforms and their relationship to cardiovascular disease progression.

- C. V., University of Minnesota** 2021–2022
Graduate Researcher
- Performed high-fidelity computational simulations of collagen gels under compression.
- C. H., University of Minnesota** 2020–2023
Undergraduate Researcher
- Developed image-based models through segmentation and computational simulation of Marfan syndrome mouse vasculature.
 - Subsequently admitted to medical school at the University of Minnesota.

Professional Service and Outreach

- Mentor, Science Small Groups (SSG), Stanford University** 2026–Present
Community College Research Mentoring Program
- Mentor community college students in developing independent research questions, formulating hypotheses, designing experimental approaches, and evaluating scientific literature.
- Ad Hoc Peer Reviewer** 2024–Present
Cardiovascular Engineering and Technology; Discover Nano; Scientific Reports
- CEMS Diversity, Equity, and Inclusion Working Group, University of Minnesota** 2020–2022
Co-founder and Chair, Graduate Recruitment Subgroup
- Contributed to the strategic plan, coordinated recruitment initiatives, and communicated with departmental and college leadership.
- Graduate Student Coordinator, University of Minnesota** 2021–2022
College of Science and Engineering
- Supported programming, recruitment, retention, and inclusion initiatives for women and BIPOC graduate students and prepared annual impact recommendations.

Selected Publications

Peer-Reviewed Journal Publications

1. Dwivedi KK, Wu Y, **Bazzi MS**, et al. Alterations in ascending aortic hemodynamics and aortic length correlate with sex-specific thoracic aortic aneurysm dilation and lifespan in a mouse model of severe Marfan syndrome. *Computers in Biology and Medicine*, 205:111594, 2026.
2. Wu Y, Dwivedi KK, Wiputra H, **Bazzi MS**, et al. Patient-specific fluid–structure interaction simulations suggest wall-shear-stress-related biomarkers in type B dissection associated with Marfan syndrome. *Annals of Biomedical Engineering*, 2026.
3. Ketterling JA, Wahyulaksana G, **Bazzi MS**, Wiputra H. A technique to track scatterers for continuous high-speed plane-wave ultrasound simulations based on a fluid domain model. *IEEE Open Journal of Ultrasonics, Ferroelectrics, and Frequency Control*, 5:166–175, 2025. **DOI**
4. **Bazzi MS**, Wiputra H, Barocas VH. Fluid–structure-based multiscale model of hemodynamics-driven arterial growth and remodeling. *Biomechanics and Modeling in Mechanobiology*, 2025. **DOI**
5. **Bazzi MS**, Wiputra H, Wood DK, Barocas VH. Computational analysis of flow and transport suggests reduced oxygen levels within intracranial aneurysms, especially in individuals with sickle-cell disease. *Journal of Biomechanical Engineering*, 147(2), 2025.
6. Komosa ER, Lin W-H, Mahadik B, **Bazzi M**, Townsend D, Fisher J, Ogle BM. A novel perfusion bioreactor promotes the expansion of pluripotent stem cells in a 3D-bioprinted chamber. *Biofabrication*, 16(1):014101, 2023.
7. Vidmar CS, **Bazzi M**, Lai VK. Computational and experimental comparison on the effects of flow-induced compression on the permeability of collagen gels. *Journal of the Mechanical Behavior of Biomedical Materials*, 2022.
8. **Bazzi MS**, Balouchzadeh R, Pavey SN, Quirk JD, Yanagisawa H, Vedula V, Wagenseil JE, Barocas VH. Mouse-specific biofluidic and geometric biomarkers of ascending thoracic aortic aneurysm outcomes. *Cardiovascular Engineering and Technology*, 2022.

9. **Bazzi MS**, Valdez JM, Barocas VH, Wood DK. A combined experimental–computational approach to quantify patient-specific blood rheology predicts altered systemic blood flow in sickle cell disease. *Biophysical Journal*, 2020.
10. **Bazzi MS**, Carvalho MS. Effect of viscoelastic forces on liquid sheet rupture process. *Journal of Non-Newtonian Fluid Mechanics*, 2019.

Preprints and Submitted Manuscripts

1. **Bazzi MS** et al. Experimental–computational characterization of somatic growth of the inferior vena cava in juvenile sheep. *Submitted*.
2. **Bazzi MS**, Adil D, Humphrey JD, Marsden AL. Data-driven discovery of constitutive models for vein mechanics using constitutive neural networks. *Submitted*.
3. Matsuzaki Y, Lindsey S, Szafron JM, Zbinden JC, Ramachandra AB, Reinhardt JW, Blum KM, **Bazzi MS**, et al. Improving tissue engineered vascular graft performance using an experimental–computational approach. *Submitted*.

Manuscripts in Preparation

1. **Bazzi MS** et al. Temporal evolution of hemodynamics in tissue-engineered vascular grafts: A fluid–structure interaction analysis. *In preparation*.
2. **Bazzi MS** et al. Mapping scaffold properties to remodeling trajectories in tissue-engineered vascular grafts. *In preparation*.

Book Chapters

1. **Bazzi MS**, Balbino TA. Dynamic microphysiological systems to access sickle cell disease: A case study for disease modeling. In: *Smart Organ-on-Chip Devices: Dynamic Microfluidic Systems for Cell Culture*, pp. 143–156, Academic Press, 2025. [DOI](#)

Invited Talks

15. **Bazzi MS**. Predicting TEVG outcomes: How scaffold design shapes vascular remodeling and blood flow. *Rising Stars in Computational and Data Sciences Workshop, Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin*, April 2025.
14. **Bazzi MS**. Multiscale, multiphysical modeling of blood flow, vascular mechanics, and growth and remodeling. *Invited Seminar, Department of Mechanical and Aerospace Engineering, University of California San Diego, San Diego, CA*, April 2025.
13. **Bazzi MS**. Subject-specific computational models of FblnSMKO+ mice: Investigating potential biomarkers to predict ascending thoracic aortic aneurysm outcomes. *CVI Early Career Research Roundtable, Stanford University*, March, 2024.
12. **Bazzi MS**. Multiscale, multiphysical phenomena in arterial remodeling. *Invited Virtual Seminar, Brazilian Association of Engineering and Mechanical Sciences*, December, 2021.
11. **Bazzi MS**, Barocas VH. Multiscale, multiphysical phenomena in arterial remodeling. *Invited Virtual Seminar, Pontifical Catholic University of Rio de Janeiro*, November, 2021.

Selected Conference Presentations

- P12. **Bazzi MS**. Microstructure-based constitutive model discovery for venous mechanics. *Oral Presentation, 8th International Conference on Computational and Mathematical Biomedical Engineering (CMBE 2026)*, Kobe, Japan, 2026.
- P11. **Bazzi MS**. Linking Scaffold Mechanics to Hemodynamics and Neovessel Remodeling in a TEVG Ovine Model. *10th International Biofluid Mechanics and Mechanobiology Symposium (IBMS10)*, Irvine, California, 2025.
- P10. **Bazzi MS**, Wiputra H, Barocas VH. Uniform growth laws recapitulate aspects of ascending aortic aneurysm progression in FBLN4 mice. *Oral Presentation, Summer Biomechanics, Bioengineering, and Biotransport Conference*, Lake Geneva, WI, June 2024.

- P9. **Bazzi MS**, Wiputra H, Barocas VH. Impact of sickle cell disease on oxygen transport in intracranial aneurysms. *Oral Presentation, 28th Congress of the European Society of Biomechanics*, Maastricht, Netherlands, July 2023.
- P8. **Bazzi MS**, Wagenseil JE, Barocas VH. Hemodynamics-driven aortic growth and remodeling in genetically modified mouse models. *Oral Presentation, 27th Congress of the European Society of Biomechanics*, Porto, Portugal, July 2022.
- P7. **Bazzi MS**, Wagenseil JE, Barocas VH. Fluid–structure-based computational modeling of hemodynamics-driven aortic growth and remodeling. *Oral Presentation, Summer Biomechanics, Bioengineering, and Biotransport Conference*, Cambridge, MN, June 2022.
- P6. **Bazzi MS**, Valdez JM, Barocas VH, Wood DK. Sickle-cell-disease-induced changes in blood viscosity can alter arterial wall shear stress profiles. *Oral Presentation, Summer Biomechanics, Bioengineering, and Biotransport Conference*, Virtual, [2021].
- P5. **Bazzi MS**, Valdez JM, Barocas VH, Wood DK. Combined microfluidic–computational approach to quantify the effect of sickle-cell disease on blood rheology. *Oral Presentation, Summer Biomechanics, Bioengineering, and Biotransport Conference*, Seven Springs, PA, [2019].
- P4. **Bazzi MS**, Carvalho MS. Breakup of viscoelastic liquid curtains. *Oral Presentation, 19th International Coating Science and Technology Symposium*, Long Beach, CA, 2018.
- P3. **Bazzi MS**, Carvalho MS. Numerical investigation of viscoelastic liquid curtain breakup. *Oral Presentation, 24th ABCM International Congress of Mechanical Engineering*, Curitiba, Brazil, 2017.
- P2. **Bazzi MS**, Carvalho MS. Effect of viscoelasticity on liquid curtain rupture. *89th Annual Meeting of the Society of Rheology*, Denver, CO, 2017.
- P1. **Bazzi MS**, Carvalho MS. Modeling liquid curtain rupture. *Coating Process Fundamentals Program, IPRIME Annual Review*, University of Minnesota, Minneapolis, MN, 2017.